

HW2

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Q1

Suppose that $\underline{y}$ is an $n \times 1$ observable random vector that follows Model II.

Note: My answers rely on the row-space criterion for estimability (Theorem 2.1): under Models II–VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $\underline{\lambda}' \in \mathcal{R}(X)$, where $\mathcal{R}(X) = \{\underline{a}'X : \underline{a} \in \mathbb{R}^n\}$.

(a)

If $\underline{\lambda}'_1\underline{\beta}$ and $\underline{\lambda}'_2\underline{\beta}$ are both estimable, is $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta}$ necessarily estimable? Explain your answer.

Answer. Yes.

Since $\underline{\lambda}'_1\underline{\beta}$ and $\underline{\lambda}'_2\underline{\beta}$ are estimable, Theorem 2.1 gives $n \times 1$ vectors $\underline{a}_1$ and $\underline{a}_2$ with $\underline{\lambda}'_1 = \underline{a}'_1X$ and $\underline{\lambda}'_2 = \underline{a}'_2X$. Then

$$(\underline{\lambda}_1 + \underline{\lambda}_2)' = (\underline{a}_1 + \underline{a}_2)'X \in \mathcal{R}(X),$$

so $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta} = (\underline{\lambda}_1 + \underline{\lambda}_2)'\underline{\beta}$ is estimable by Theorem 2.1.

(Said perhaps more succinctly: If both $\underline{\lambda}'_1\underline{\beta}$ and $\underline{\lambda}'_2\underline{\beta}$ are estimable, then $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta}$ is itself a linear combination of two estimable quantities, and is hence estimable.)

(b)

If $\underline{\lambda}'_1\underline{\beta}$ and $\underline{\lambda}'_2\underline{\beta}$ are both nonestimable, is $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta}$ necessarily nonestimable? Explain your answer.

Answer. No.

Let $\underline{\lambda}'_1\underline{\beta}$ be nonestimable, so $\underline{\lambda}'_1 \notin \mathcal{R}(X)$ by Theorem 2.1, and set $\underline{\lambda}_2 = -\underline{\lambda}_1$. Since $\mathcal{R}(X)$ is a subspace, $\underline{\lambda}'_2 = -\underline{\lambda}'_1 \notin \mathcal{R}(X)$ as well, so $\underline{\lambda}'_2\underline{\beta}$ is also nonestimable.

However, by adding a “clever zero”,

$$\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta} = \underline{\lambda}'_1\underline{\beta} + (-\underline{\lambda}'_1\underline{\beta}) = \underline{0}'_p\underline{\beta} = 0,$$

which *is* estimable, as $\underline{0}'_p = \underline{0}'_nX \in \mathcal{R}(X)$.

Hence the sum of two nonestimable functions can be estimable, so $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta}$ is not necessarily nonestimable.

Note: This answer requires $\text{rank}(X) < p$; otherwise $\mathcal{R}(X) = \mathbb{R}^p$ and every $\underline{\lambda}'\underline{\beta}$ is estimable.

(c)

Can $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta}$ be estimable if $\underline{\lambda}'_1\underline{\beta}$ is estimable and $\underline{\lambda}'_2\underline{\beta}$ is nonestimable? Explain your answer.

Answer. No ($\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta}$ would necessarily be nonestimable).

Suppose, for contradiction, that $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta}$ is estimable. By Theorem 2.1, $\underline{\lambda}'_1 + \underline{\lambda}'_2 \in \mathcal{R}(X)$ and $\underline{\lambda}'_1 \in \mathcal{R}(X)$, and since $\mathcal{R}(X)$ is a subspace,

$$\underline{\lambda}'_2 = (\underline{\lambda}'_1 + \underline{\lambda}'_2) - \underline{\lambda}'_1 \in \mathcal{R}(X).$$

So $\underline{\lambda}'_2 \underline{\beta}$ is estimable by Theorem 2.1, contradicting the assumption that it is nonestimable.

Q2

Take the model to be Model VII and suppose that

$$\underline{y} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \quad \text{and} \quad X = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

(a)

Find the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and on $\mathcal{C}^\perp(X)$.

Answer. The projections are $(2, 3, 0)'$ on $\mathcal{C}(X)$ and $(0, 0, 4)'$ on $\mathcal{C}^\perp(X)$.

To find these projections, we utilize the following results/definitions:

- $\mathcal{C}(X) = \{X\underline{b} : \underline{b} \in \mathbb{R}^3\}$ is the span of the columns of X ,
- $\mathcal{C}^\perp(X) = \{\underline{a} \in \mathbb{R}^3 : X'\underline{a} = \underline{0}\}$ is the null space of X' , and
- **Theorem 2.6:** Any $n \times 1$ vector $\underline{y}$ admits a unique decomposition $\underline{y} = \underline{y}_1 + \underline{y}_2$ with $\underline{y}_1 \in \mathcal{C}(X)$ and $\underline{y}_2 \in \mathcal{C}^\perp(X)$.
 - What's more, this decomposition is given by $\underline{y}_1 = X\widehat{\underline{\beta}}$ and $\underline{y}_2 = \underline{y} - X\widehat{\underline{\beta}}$ for any solution $\widehat{\underline{\beta}}$ to the normal equations.
 - And the vectors $\underline{y}_1$ and $\underline{y}_2$ are the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and $\mathcal{C}^\perp(X)$, resp.

We'll next detail the computation of the projections.

$\mathcal{C}(X)$

The columns of X as given are

$$\underline{x}_1 = (1, 2, 0)', \quad \underline{x}_2 = (2, 3, 0)', \quad \underline{x}_3 = (3, 1, 0)'.$$

Each has a shared third coordinate, 0, such that $\mathcal{C}(X) \subseteq \{(u, v, 0)' : u, v \in \mathbb{R}\}$.

Also note that the vectors

$$-3\underline{x}_1 + 2\underline{x}_2 = (1, 0, 0)' \quad \text{and} \quad 2\underline{x}_1 - \underline{x}_2 = (0, 1, 0)'$$

both lie in $\mathcal{C}(X)$, and every $(u, v, 0)'$ is a linear combination of these two.

Hence

$$\mathcal{C}(X) = \{(u, v, 0)' : u, v \in \mathbb{R}\}.$$

$\mathcal{C}^\perp(X)$

For $\underline{a} = (a_1, a_2, a_3)'$,

$$X'\underline{a} = (a_1 + 2a_2, 2a_1 + 3a_2, 3a_1 + a_2)'.$$

Since $\det \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = -1 \neq 0$, the first two coordinates vanish only if $a_1 = a_2 = 0$. The third then vanishes as well, and a_3 is unrestricted. Hence

$$\mathcal{C}^\perp(X) = \{(0, 0, w)' : w \in \mathbb{R}\}.$$

Combining it all together.

Splitting $\underline{y}$,

$$\underline{y} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix},$$

with the first vector in $\mathcal{C}(X)$ and the second in $\mathcal{C}^\perp(X)$.

By the uniqueness in Theorem 2.6, these are the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and $\mathcal{C}^\perp(X)$, resp.

(b)

Determine $\text{rank}(X'X)$.

Answer. $\text{rank}(X'X) = 2$.

Note: By **Lemma 2.3**, $\text{rank}(A'A) = \text{rank}(A)$ for any matrix A .

From part (a), $\mathcal{C}(X)$ is spanned by the two linearly independent vectors $(1, 0, 0)'$ and $(0, 1, 0)'$, so $\text{rank}(X) = \dim \mathcal{C}(X) = 2$.

Via Lemma 2.3, with $A = X$, we then have

$$\text{rank}(X'X) = \text{rank}(X) = 2.$$

(c)

Use the following method and find a generalized inverse of $X'X$.

Let A be an $n \times m$ matrix of rank $r \geq 1$. Let B be a nonsingular $r \times r$ submatrix of A . For matrix A , replace the entries of B by those of $(B^{-1})'$, and replace all other entries of A by 0. Then the transpose of the resulting matrix is a generalized inverse of A .

Answer.

By definition, a *generalized inverse* of a matrix A is any matrix G satisfying $AGA = A$.

Here, we have:

$$A = X'X = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 5 & 8 & 5 \\ 8 & 13 & 9 \\ 5 & 9 & 10 \end{pmatrix},$$

which has rank $r = 2$ by part (b).

Then

$$B = \begin{pmatrix} 5 & 8 \\ 8 & 13 \end{pmatrix}, \quad \det(B) = 65 - 64 = 1 \neq 0, \quad (B^{-1})' = B^{-1} = \begin{pmatrix} 13 & -8 \\ -8 & 5 \end{pmatrix}.$$

Utilizing the method noted (replacing the entries of B in A by those of $(B^{-1})'$, and all other entries by 0), we have

$$G = (X'X)^- = \begin{pmatrix} 13 & -8 & 0 \\ -8 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

And since B is symmetric, the resulting matrix is symmetric, so its transpose (the final step of the method) is equal to itself.

We then verify this is in fact a generalized inverse:

$$AG = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -7 & 5 & 0 \end{pmatrix} \Rightarrow AGA = \begin{pmatrix} 5 & 8 & 5 \\ 8 & 13 & 9 \\ 5 & 9 & 10 \end{pmatrix} = A.$$

So G is a generalized inverse of $X'X$, as desired.

(d)

Compute a solution $\hat{\underline{\beta}}$ to the normal equations.

Answer. $\hat{\underline{\beta}} = (0, 1, 0)'$.

To compute this solution, we utilize the following results:

- **Theorem 2.3:** The normal equations $X'X\hat{\underline{\beta}} = X'\underline{y}$ are consistent.
- **Theorem 1.2:** A matrix G is a generalized inverse of a matrix A if and only if $\underline{w} = G\underline{c}$ solves $A\underline{w} = \underline{c}$ for every vector $\underline{c}$ for which these equations are consistent.

With these results in hand, since the normal equations are consistent (Theorem 2.3), Theorem 1.2 with $A = X'X$ and $\underline{c} = X'\underline{y}$ gives that $\hat{\underline{\beta}} = (X'X)^- X'\underline{y}$ solves them.

Since

$$X'\underline{y} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 13 \\ 9 \end{pmatrix},$$

taking $(X'X)^- = G$ from part (c) gives

$$\hat{\underline{\beta}} = \begin{pmatrix} 13 & -8 & 0 \\ -8 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 8 \\ 13 \\ 9 \end{pmatrix} = \begin{pmatrix} 104 - 104 \\ -64 + 65 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

(e)

Check that the vectors $X\hat{\underline{\beta}}$ and $\underline{y} - X\hat{\underline{\beta}}$ agree with your results in part (a).

Answer.

From part (d), $X\hat{\underline{\beta}}$ is the second column of X , so

$$X\hat{\underline{\beta}} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}, \quad \underline{y} - X\hat{\underline{\beta}} = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix},$$

which match the projections in part (a), as Theorem 2.6 requires.

A quick numerical check of parts (c)–(e) in R:

```
X <- matrix(c(1, 2, 3,
              2, 3, 1,
              0, 0, 0),
            nrow = 3,
            byrow = TRUE)
```

```

y <- c(2, 3, 4)
XtX <- crossprod(X)
G <- matrix(c(13, -8, 0,
              -8, 5, 0,
              0, 0, 0),
            nrow = 3,
            byrow = TRUE)

# verify G is a generalized inverse of X'X
XtX %*% G %*% XtX

```

```

##      [,1] [,2] [,3]
## [1,]    5    8    5
## [2,]    8   13    9
## [3,]    5    9   10

```

```
XtX
```

```

##      [,1] [,2] [,3]
## [1,]    5    8    5
## [2,]    8   13    9
## [3,]    5    9   10

```

```

bhat <- G %*% crossprod(X, y)
# bhat
# Expect (0, 1, 0)
bhat

```

```

##      [,1]
## [1,]    0
## [2,]    1
## [3,]    0

```

```

# X bhat
# Expect (2, 3, 0)
X %*% bhat

```

```

##      [,1]
## [1,]    2
## [2,]    3
## [3,]    0

```

```

# y - X bhat
# Expect (0, 0, 4)
y - X %*% bhat

```

```

##      [,1]
## [1,]    0
## [2,]    0
## [3,]    4

```

Q3

For Model VII, partition $\underline{y}$, X , and $\underline{\epsilon}$ correspondingly as follows:

$$\underline{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \quad \text{and} \quad \underline{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \end{pmatrix}.$$

Suppose that the row spaces of X_1 and X_2 are essentially disjoint, that is, $\mathcal{R}(X_1) \cap \mathcal{R}(X_2) = \{0'\}$. Show that, if $\underline{\lambda}' \in \mathcal{R}(X_1)$, then the least squares estimate of $\underline{\lambda}'\underline{\beta}$ based on the model $\underline{y}_1 = X_1\underline{\beta} + \underline{\epsilon}_1$ is the same as the least squares estimate of $\underline{\lambda}'\underline{\beta}$ based on the model $\underline{y} = X\underline{\beta} + \underline{\epsilon}$.

Hint: Under the full model, the LSE is $\underline{a}'X'\underline{y} = \underline{a}'X'_1\underline{y}_1 + \underline{a}'X'_2\underline{y}_2$ and, under the partial model, the LSE is $\underline{b}'X'_1\underline{y}_1$ where here $X'_1X_1\underline{a} + X'_2X_2\underline{a} = X'X\underline{a} = \underline{\lambda} = X'_1X_1\underline{b}$ (why?); in the last equation, move terms involving X'_2 and X'_1 to different sides.

Answer.

Note: X_1 is $n_1 \times p$ and X_2 is $n_2 \times p$, with $n_1 + n_2 = n$. The partial model $\underline{y}_1 = X_1\underline{\beta} + \underline{\epsilon}_1$ is itself an instance of Model VII, as $E(\underline{\epsilon}_1) = \underline{0}_{n_1}$ and $\text{Var}(\underline{\epsilon}_1)$ is the leading $n_1 \times n_1$ block of $\text{Var}(\underline{\epsilon})$. This means we can apply key (known) results to both models based on X_1 and X , resp., in this problem.

That said, the key results we will use are as follows:

- **Theorem 2.5:** Under Models II–VII, $\underline{\lambda}'\underline{\hat{\beta}} = \underline{\lambda}'\underline{\tilde{\beta}}$ for any two solutions $\underline{\hat{\beta}}$ and $\underline{\tilde{\beta}}$ to the normal equations if and only if $\underline{\lambda}'\underline{\beta}$ is estimable.
- **Theorem 2.7:** Under Models II–VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if the conjugate normal equations $X'X\underline{r} = \underline{\lambda}$ are consistent.
- **Theorem 2.8:** Under Models II–VII, the least squares estimate of an estimable $\underline{\lambda}'\underline{\beta}$ equals $\underline{r}'X'\underline{y}$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$.
- **Lemma 1.3(ii):** For any matrices M , P , and Q of appropriate dimensions, $M'MP = M'MQ$ implies $MP = MQ$.

Since $\underline{\lambda}' \in \mathcal{R}(X_1)$, $\underline{\lambda}' = \underline{t}'X_1$ for some $n_1 \times 1$ vector $\underline{t}$, and so

$$\underline{\lambda}' = (\underline{t}', \underline{0}'_{n_2}) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \in \mathcal{R}(X)$$

as well. So $\underline{\lambda}'\underline{\beta}$ is estimable under both models (Theorem 2.1, noted in Q1), and by Theorem 2.5 its least squares estimate is well defined under each. By Theorem 2.7, estimability makes the conjugate normal equations consistent under each model, so there are $p \times 1$ vectors $\underline{r}$ and $\underline{r}_1$ (the hint's $\underline{a}$ and $\underline{b}$) with

$$X'X\underline{r} = \underline{\lambda} \quad \text{and} \quad X'_1X_1\underline{r}_1 = \underline{\lambda},$$

and by Theorem 2.8 the least squares estimates under the full and partial models are $\underline{r}'X'\underline{y}$ and $\underline{r}'_1X'_1\underline{y}_1$, resp.

By block multiplication, $X'X = X'_1X_1 + X'_2X_2$ and $X'\underline{y} = X'_1\underline{y}_1 + X'_2\underline{y}_2$, so

$$X'_1X_1\underline{r} + X'_2X_2\underline{r} = \underline{\lambda} = X'_1X_1\underline{r}_1, \quad \text{that is,} \quad X'_1X_1(\underline{r} - \underline{r}_1) = -X'_2X_2\underline{r}.$$

Transposing both sides gives

$$[X_1(\underline{r} - \underline{r}_1)]'X_1 = [-X_2\underline{r}]'X_2,$$

where the left side is of the form $\underline{u}'X_1$, and so lies in $\mathcal{R}(X_1)$; likewise, the right side lies in $\mathcal{R}(X_2)$. Since $\mathcal{R}(X_1) \cap \mathcal{R}(X_2) = \{0'\}$, both sides equal $\underline{0}'_p$. Transposing back,

$$X'_1X_1\underline{r} = X'_1X_1\underline{r}_1 \quad \text{and} \quad X'_2X_2\underline{r} = X'_2X_2\underline{0}_p.$$

By Lemma 1.3(ii) ($M'MP = M'MQ$ implies $MP = MQ$), $X_1\underline{r} = X_1\underline{r}_1$ and $X_2\underline{r} = \underline{0}_{n_2}$. Therefore

$$\underline{r}'X'\underline{y} = (X_1\underline{r})'\underline{y}_1 + (X_2\underline{r})'\underline{y}_2 = (X_1\underline{r}_1)'\underline{y}_1 = \underline{r}'_1X'_1\underline{y}_1,$$

so the least squares estimates of $\underline{\lambda}'\underline{\beta}$ under the two models coincide.

Q4

Show that, under Model II, $\underline{a}'\underline{y}$ is a BLUE of its expectation if and only if $\underline{a} \in \mathcal{C}(X)$.

Answer.

Note: Under Model II, we have:

- $\underline{a}$ is $n \times 1$,
- $E(\underline{\epsilon}) = \underline{0}_n$, and
- $\text{Var}(\underline{y}) = \text{Var}(\underline{\epsilon}) = \sigma^2 I_n$ with $\sigma^2 > 0$.

We also utilize the following results:

- **Lemma 1.4:** For any matrix X , $X'X(X'X)^{-}X' = X'$.
- **Corollary 1(ii) to Lemma 1.4:** For any matrix X , $X(X'X)^{-}X'X = X$.
- **Corollary 3 to Lemma 1.5:** For any matrix X , $X(X'X)^{-}X'$ is symmetric.
- **Corollary to Lemma 2.4:** For a random vector $\underline{x}$ with $\text{Var}(\underline{x}) = \Sigma$ and a matrix of constants L , $\text{Var}(L'\underline{x}) = L'\Sigma L$.
- **Theorem 2.9 (Gauss–Markov):** Under Models II–VII, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is any solution to the normal equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. What's more, under Model II, the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is no greater than that of any other linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, for every $\sigma^2 > 0$; said differently, $\underline{\lambda}'\widehat{\underline{\beta}}$ is the BLUE.

We also utilize the properties of the projection matrix, $P_X = X(X'X)^{-}X'$, an $n \times n$ matrix that satisfies:

- $P_X X = X$,
- $P_X' = P_X$, and
- $P_X^2 = X(X'X)^{-}[X'X(X'X)^{-}X'] = X(X'X)^{-}X' = P_X$.

It then follows that $P_X'(I_n - P_X) = P_X - P_X^2 = \underline{0}_{n \times n}$.

Taking this information as given, then, we can proceed with the proof.

Proof:

Let $\underline{\lambda}' = \underline{a}'X$, so that $E(\underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$ and $\underline{a}'\underline{y}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, which is estimable by Theorem 2.1 (noted in Q1). By Theorems 2.3 and 1.2 (noted in Q2(d)), $\widehat{\underline{\beta}} = (X'X)^{-}X'\underline{y}$ solves the normal equations, so by the Gauss–Markov theorem (Theorem 2.9), the BLUE of $\underline{\lambda}'\underline{\beta}$ is

$$\underline{\lambda}'\widehat{\underline{\beta}} = \underline{a}'X(X'X)^{-}X'\underline{y} = \underline{a}'P_X\underline{y} = (P_X\underline{a})'\underline{y}.$$

Now write $\underline{a} = P_X\underline{a} + \underline{e}$, where $\underline{e} \equiv (I_n - P_X)\underline{a}$. The cross term $\underline{a}'P_X'\underline{e} = \underline{a}'P_X'(I_n - P_X)\underline{a}$ vanishes, so by the Corollary to Lemma 2.4 ($\text{Var}(L'\underline{y}) = L'\text{Var}(\underline{y})L$),

$$\text{Var}(\underline{a}'\underline{y}) = \sigma^2 \underline{a}'\underline{a} = \sigma^2 (P_X\underline{a})'(P_X\underline{a}) + \sigma^2 \underline{e}'\underline{e} = \text{Var}((P_X\underline{a})'\underline{y}) + \sigma^2 \underline{e}'\underline{e}.$$

Since $\sigma^2 > 0$, the linear unbiased estimator $\underline{a}'\underline{y}$ attains the BLUE's variance, and so is itself a BLUE, if and only if $\underline{e}'\underline{e} = 0$. As a sum of squares, $\underline{e}'\underline{e} = 0$ if and only if $\underline{e} = \underline{0}_n$, that is, $\underline{a} = P_X\underline{a}$.

It remains to show $\underline{a} = P_X\underline{a}$ if and only if $\underline{a} \in \mathcal{C}(X)$:

- ($\Rightarrow$) If $\underline{a} = P_X\underline{a}$, then $\underline{a} = X\underline{b}$ with $\underline{b} = (X'X)^{-}X'\underline{a}$, so $\underline{a} \in \mathcal{C}(X)$.
- ($\Leftarrow$) If $\underline{a} = X\underline{b}$ for some $\underline{b} \in \mathbb{R}^p$, then $P_X\underline{a} = P_X X\underline{b} = X\underline{b} = \underline{a}$.

Hence $\underline{a}'\underline{y}$ is a BLUE of its expectation if and only if $\underline{a} \in \mathcal{C}(X)$.

Q5

(a)

Show that, under Model II, the problem of determining the BLUE $\underline{a}'\underline{y}$ of an estimable function $\underline{\lambda}'\underline{\beta}$ is equivalent to that of minimizing $\underline{a}'\underline{a}$ subject to the constraints $\underline{a}'X = \underline{\lambda}'$.

Answer.

Note: $\underline{a}$ is $n \times 1$ and $\underline{\lambda}$ is $p \times 1$.

By Lemma 2.1, a linear estimator $c + \underline{a}'\underline{y}$ is unbiased for $\underline{\lambda}'\underline{\beta}$ if and only if $c = 0$ and $\underline{a}'X = \underline{\lambda}'$.

So the linear unbiased estimators of $\underline{\lambda}'\underline{\beta}$ are exactly

$$\{\underline{a}'\underline{y} : \underline{a}'X = \underline{\lambda}'\},$$

which is nonempty because $\underline{\lambda}'\underline{\beta}$ is estimable.

Under Model II, $\text{Var}(\underline{y}) = \sigma^2 I_n$, so by the Corollary to Lemma 2.4 (noted in Q4), each has variance

$$\text{Var}(\underline{a}'\underline{y}) = \underline{a}'(\sigma^2 I_n)\underline{a} = \sigma^2 \underline{a}'\underline{a}.$$

Since $\sigma^2 > 0$ does not depend on $\underline{a}$, it then follows that $\underline{a}$ minimizes $\text{Var}(\underline{a}'\underline{y})$ over this class if and only if it minimizes $\underline{a}'\underline{a}$ subject to $\underline{a}'X = \underline{\lambda}'$ (equivalently, $X'\underline{a} = \underline{\lambda}$).

And since the minimizing $\underline{a}$ does not depend on σ^2 , the resulting estimator has minimum variance uniformly in σ^2 .

(b)

The Lagrangian function associated with this minimization problem is

$$L(\underline{a}, \underline{\rho}) = \underline{a}'\underline{a} + 2\underline{\rho}'(X'\underline{a} - \underline{\lambda}).$$

Use the vector calculus to differentiate the Lagrangian function with respect to $\underline{a}$ and the Lagrangian multipliers, and set these derivatives equal to zero.

Answer.

Note: There is one multiplier per constraint in $X'\underline{a} = \underline{\lambda}$, so $\underline{\rho}$ is $p \times 1$.

We utilize the following identities, for a vector variable $\underline{w}$ and a constant vector $\underline{h}$:

- $\partial(\underline{h}'\underline{w})/\partial\underline{w} = \underline{h}$, and
- $\partial(\underline{w}'\underline{w})/\partial\underline{w} = 2\underline{w}$.

Expanding,

$$L(\underline{a}, \underline{\rho}) = \underline{a}'\underline{a} + 2(X\underline{\rho})'\underline{a} - 2\underline{\rho}'\underline{\lambda} = \underline{a}'\underline{a} + 2(X'\underline{a} - \underline{\lambda})'\underline{\rho}.$$

By these identities, the first form gives the gradient in $\underline{a}$ and the second the gradient in $\underline{\rho}$:

$$\frac{\partial L}{\partial \underline{a}} = 2\underline{a} + 2X\underline{\rho}, \quad \frac{\partial L}{\partial \underline{\rho}} = 2(X'\underline{a} - \underline{\lambda}).$$

Setting these equal to zero gives the system

$$\underline{a} + X\underline{\rho} = \underline{0}_n, \quad X'\underline{a} = \underline{\lambda}. \quad (\text{Q5 (b)})$$

(c)

Show that the resulting equations have a solution if and only if the conjugate normal equations are consistent.

Answer.

($\Rightarrow$) Suppose $(\underline{a}_0, \underline{\rho}_0)$ solves (Q5 (b)). The first equation gives $\underline{a}_0 = -X\underline{\rho}_0$, and substituting into the second,

$$\underline{\lambda} = X'\underline{a}_0 = X'X(-\underline{\rho}_0).$$

So $\underline{r} = -\underline{\rho}_0$ solves $X'X\underline{r} = \underline{\lambda}$, and the conjugate normal equations are consistent.

($\Leftarrow$) Suppose $X'X\underline{r}_0 = \underline{\lambda}$ for some $p \times 1$ vector $\underline{r}_0$. Set $\underline{a}_0 = X\underline{r}_0$ and $\underline{\rho}_0 = -\underline{r}_0$. Then

$$\underline{a}_0 + X\underline{\rho}_0 = X\underline{r}_0 - X\underline{r}_0 = \underline{0}_n \quad \text{and} \quad X'\underline{a}_0 = X'X\underline{r}_0 = \underline{\lambda},$$

so $(\underline{a}_0, \underline{\rho}_0)$ solves (Q5 (b)).

(d)

Solve your equations for the minimizing value of $\underline{a}$.

Answer. $\underline{a}_* = X(X'X)^{-}\underline{\lambda}$.

We utilize the following results, each noted earlier: Theorem 1.2 (Q2(d)), Theorem 2.7 (Q3), and Lemma 1.3(ii) (Q3).

Since $\underline{\lambda}'\underline{\beta}$ is estimable, the conjugate normal equations are consistent (Theorem 2.7, noted in Q3), so (Q5 (b)) has a solution by part (c). For any solution $(\underline{a}, \underline{\rho})$, the first equation gives $\underline{a} = X\underline{r}$ with $\underline{r} = -\underline{\rho}$, and substituting into the second gives $X'X\underline{r} = \underline{\lambda}$. By Theorem 1.2 (noted in Q2(d)), $(X'X)^{-}\underline{\lambda}$ also solves $X'X\underline{r} = \underline{\lambda}$, so $X'X\underline{r} = X'X(X'X)^{-}\underline{\lambda}$, and Lemma 1.3(ii) (noted in Q3) gives

$$\underline{a} = X\underline{r} = X(X'X)^{-}\underline{\lambda} \equiv \underline{a}_*.$$

So every solution of (Q5 (b)) has the same $\underline{a}$ -component, $\underline{a}_*$.

Since (Q5 (b)) gives necessary conditions only, we verify directly that $\underline{a}_*$ minimizes $\underline{a}'\underline{a}$ subject to $X'\underline{a} = \underline{\lambda}$. Let $\underline{a}$ be any $n \times 1$ vector with $X'\underline{a} = \underline{\lambda}$, and set $\underline{d} = \underline{a} - \underline{a}_*$, so that $X'\underline{d} = \underline{0}_p$. Since $\underline{a}_* = X\underline{r}_*$ with $\underline{r}_* = (X'X)^{-}\underline{\lambda}$, we have $\underline{a}_*\underline{d} = \underline{r}_*X'\underline{d} = 0$, and therefore

$$\underline{a}'\underline{a} = (\underline{a}_* + \underline{d})'(\underline{a}_* + \underline{d}) = \underline{a}_*\underline{a}_* + \underline{d}'\underline{d} \geq \underline{a}_*\underline{a}_*,$$

with equality if and only if $\underline{d} = \underline{0}_n$. So $\underline{a}_*$ is the unique minimizing value of $\underline{a}$.